

## DIGITAL FORENSIC INVESTIGATION REPORT

Deleted data recovery and activity timeframe review

**Prepared for: Management, Apex Integrated Technologies Ltd.**

| CASE DETAILS               | VALUE                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------|
| Case ID in assignment      | AIT-DF-2026-009                                                                               |
| Investigation type         | Digital forensics / deleted data recovery                                                     |
| Primary replacement image  | Omonoa Rerun.001–.004 (split image set)                                                       |
| Additional image reference | Evidence.E01 appears in screenshots and hash list; file not present in NIIT folder            |
| Prepared                   | 27 September 2026                                                                             |
| Investigator               | Omonoa Awoteinm Darego                                                                        |
| Team                       | N/A                                                                                           |
| Report status              | Preliminary findings from supplied image records, Autopsy case data, exports, and screenshots |

I prepared this report from the replacement-image materials in the NIIT folder and the case assignment. I distinguish image-acquisition records, Autopsy observations, file-export checks, and unresolved limitations throughout.

## 1. Executive Summary

I examined the replacement evidence materials supplied in the NIIT folder. They include four image segments named Omonoa Rerun.001 through .004, an FTK Imager acquisition summary, an Autopsy Project 2 case, CSV exports, file hash records, and screenshots documenting FTK Imager, R-Drive Image, and Autopsy activity. The four segments total 5,251,285,504 bytes. The FTK summary records matching MD5 and SHA-1 verification results for the acquired image. It also records one source sector that could not be read and was replaced with zeros in the image.

The Autopsy Project 2 screenshot and CSV export show 1,162 deleted-file records. In the reviewed export, the exact target paths SSH, Corel, and corel draw work appear with unallocated directory entries. The SSH folder has 10 PDF files. The corel draw work export contains 45 payload files: three CDR files, 37 JPG files, two PNG files, two PDFs, and one ZIP. I found no exact My pic, Our pic, or ADDS Project path in the reviewed deleted-file export or extracted root tree. Orphaned exports include several Active Directory and network-project documents, but their former parent folder is not established.

I recalculated MD5 and SHA-1 for 541 files in the extracted root tree under SSH, Corel, and corel draw work. All 541 match the corresponding entries in File Hash Value.csv. This confirms consistency with that hash list; it does not by itself prove the relationship between the hash list's Evidence.E01 source and the Omonoa Rerun image used by Autopsy.

I cannot state when deletion occurred, who performed it, whether it was intentional, or whether the absence of exact entries for three target folders means they never existed. Autopsy shows zero counts in its file-type categories, and the case database contains no ingest-job records. The folder also contains a separate Evidence.E01 reference with different image dimensions and hashes. I therefore treat the Omonoa Rerun set as the primary source for these preliminary findings and flag image identity and processing completeness for confirmation before a final conclusion.

## 2. Assignment and Scope

The case assignment asks me to examine a forensic image from a workstation after several folders became unavailable. It identifies six target folders: SSH, My pic, Our pic, ADDS Project, Corel, and Corel Draw Work. It requires deleted-file analysis, recovery, a defensible timeframe, cross-tool validation, and a report that distinguishes observations from interpretations.

For this report, I reviewed the acquisition summary, the Autopsy case database and CSV export, the file-hash CSV, the extracted folders under [root] and [orphan], and the supplied application screenshots. I did not alter the image segments. I did not conduct a fresh full ingest in FTK Imager, Autopsy, or R-Drive Image; the findings below are limited to the artifacts and outputs present in the NIIT folder and to read-only checks of the exported files.

### Materials reviewed

| Material              | What I found                              | How I used it                                              |
|-----------------------|-------------------------------------------|------------------------------------------------------------|
| Omonoa Rerun.001–.004 | Four segments; total 5,251,285,504 bytes. | Primary replacement-image identity and acquisition source. |

| Material                                   | What I found                                                        | How I used it                                                                 |
|--------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Omonoa Rerun.001.txt                       | FTK Imager 8.3.0.27 acquisition/verification summary.               | Image parameters, hashes, acquisition times, bad-sector limitation.           |
| Project 2/autopsy.db and Project 2 CSV.csv | Autopsy 4.23.1 case; 1,162 deleted-file rows.                       | Data-source identity, file-system metadata, path and allocation observations. |
| File Hash Value.csv and [root] exports     | MD5/SHA-1 records and exported folders.                             | Independent check of exported-file bytes against recorded hashes.             |
| Autopsy Screenshots 41–62                  | Screenshots of Project01, Project 2, FTK Imager, and R-Drive Image. | Corroborate displayed results and identify a separate Evidence.E01 artifact.  |
| Partition3-1-image.rdr                     | 700,416-byte R-Drive output file.                                   | Confirms an output file exists; no separate verification record was supplied. |

### 3. Evidence Identification and Integrity

#### 3.1 Primary replacement-image set

| Property               | Recorded result                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| Image segments         | Omonoa Rerun.001, .002, .003, .004                                                       |
| Combined size          | 5,251,285,504 bytes                                                                      |
| Acquisition tool       | FTK Imager 8.3.0.27                                                                      |
| Source type / device   | Physical; Callback Technologies Virtual Storage                                          |
| Source size / geometry | 5,008 MB; 10,256,417 sectors; 512 bytes per sector                                       |
| Acquisition interval   | 27 Sep 2026 16:30:25–16:31:06 WAT, per FTK summary                                       |
| MD5                    | c798146e6c2567c8dc0561cb89f6f555 — computed and report hashes match                      |
| SHA-1                  | 9951f28c9b2efbafcac4acb2b97b3bd8d3d883a2 — computed and report hashes match              |
| Source read limitation | Sector 10,256,384 could not be read; its contents were replaced with zeros in the image. |

I regard the FTK verification as support that the replacement image matches its stored/report hashes. It does not restore the unreadable source sector; that sector remains a known limitation. The acquisition

record identifies the source as a virtual storage device, which I preserve in the report rather than treating it as an identified physical workstation drive.

### 3.2 Autopsy data source

The Autopsy case is titled Project 2 and runs Autopsy 4.23.1. Its data source is labeled Omonoa Rerun.001\_1 Host. The case database records an image size of 5,251,285,504 bytes, time zone Africa/Lagos, and an NTFS file system at byte offset 8,388,608 with 4,096-byte blocks. Those values are consistent with the Omonoa Rerun replacement-image set. The database does not contain an image MD5/SHA-1/SHA-256 value, and its ingest\_jobs tables contain no job records; therefore I cannot use that database to establish which ingest modules completed.

### 3.3 Separate Evidence.E01 reference and earlier invalid copy

The screenshots and File Hash Value.csv also refer to a separate image named Evidence.E01. FTK Imager screenshots show 10,240,000 sectors and matching MD5 ae39e9a30ab5a65809eb93289938e149 and SHA-1 4462d8bcc58431744669af554ce9046fee9b16bd. The actual Evidence.E01 image file is not present in the NIIT folder. A later Autopsy screenshot shows an ingest error for that Evidence.E01 at sector offset 34, identified as a Microsoft Reserved partition.

This Evidence.E01 reference is not the same byte count or hash as the Omonoa Rerun set. It also differs from the earlier D:\Evidence.E01 file I examined, which was 10,341,301 bytes and consisted entirely of 0xFF bytes. I cannot establish from the current folder whether the Evidence.E01 screenshot is a replacement, a partition image, or a separate copy. I do not combine its records with the Omonoa Rerun results unless that relationship is documented.

## 4. Methods and Tool Activity

| Tool                         | Evidence of use in reviewed materials                                                                            | Finding / limitation                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FTK Imager 8.3.0.27          | Omonoa acquisition summary and verification screenshots; separate Evidence.E01 verification and tree screenshot. | Omonoa image hashes match the report; source had one unreadable sector zero-filled. Separate Evidence.E01 has different hashes and an Autopsy partition-ingest issue. |
| Autopsy 4.23.1               | Project 2 case/database and screenshots.                                                                         | Deleted-file view reports 1,162 entries; CSV has 1,162 rows. File-type categories show zero results; database has no ingest-job records.                              |
| R-Drive Image 7.3 Build 7314 | Screenshots show a 4.88 GB NTFS New Volume selected and “Execution completed / Image created successfully.”      | Partition3-1-image.rdr exists at 700,416 bytes. No hash/verification record ties it to the Omonoa set.                                                                |
| DD                           | No DD command log, image, or screenshot found.                                                                   | No DD activity is documented; DD was previously reported unavailable.                                                                                                 |

I did not treat a successful R-Drive image-creation dialog as verification of the Omonoa image. The R-Drive output and source selection need their own hash, settings, and verification record if they are to support the final evidence-integrity section.

## 5. Autopsy Deleted-File Findings

The Autopsy Project 2 screenshot displays 1,162 results under Deleted Files. The Project 2 CSV also contains 1,162 rows. All 1,162 rows have blank MD5 and SHA-256 fields, so the CSV is a deleted-entry inventory, not a hash-validated recovered-file manifest. Many rows show unallocated directory and metadata flags. I interpret those flags as support that the entries are deleted/unallocated records in this image, not as proof that every file's content is recoverable.

| Target path     | Autopsy export observations                                                                                                   | Exported material reviewed                                                                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SSH             | Unallocated SSH directory record and 10 unallocated PDF file entries.                                                         | 10 PDF copies under [root]\SSH; all 10 match MD5 and SHA-1 in File Hash Value.csv.                                                                                                                                       |
| Corel           | Unallocated Corel directory record and 632 descendant records; contents include 486 payload files in the extracted root tree. | 486 files under [root]\Corel; all 486 match corresponding hash-list entries. Most paths/names indicate Corel Content presets, brush assets, patterns, and starter-pack resources rather than clearly user-authored work. |
| corel draw work | Unallocated directory record and 47 descendant records; 45 payload files in the extracted root tree.                          | 3 CDR, 37 JPG, 2 PNG, 2 PDF, and 1 ZIP copies under [root]\corel draw work; all 45 match corresponding hash-list entries.                                                                                                |
| My pic          | No exact folder/path match in the reviewed deleted-file CSV or extracted [root] tree.                                         | No folder-specific export identified. This is not proof the folder never existed.                                                                                                                                        |
| Our pic         | No exact folder/path match in the reviewed deleted-file CSV or extracted [root] tree.                                         | No folder-specific export identified. This is not proof the folder never existed.                                                                                                                                        |
| ADDS Project    | No exact folder/path match in the reviewed deleted-file CSV or extracted [root] tree.                                         | Orphaned Active Directory/network project materials exist, but their former parent folder is unknown.                                                                                                                    |

The 541 exported files I checked (10 SSH, 486 Corel, and 45 corel draw work) match their recorded MD5 and SHA-1 values in File Hash Value.csv. This supports the integrity of those exported copies relative to that hash list. Because the hash list uses Evidence.E01 paths while Autopsy Project 2 analyzes Omonoa Rerun.001, I still need provenance showing that both outputs derive from the same evidence source before treating this as full cross-tool validation.

## 6. Target-Folder Interpretation

### SSH

The Autopsy export contains an unallocated SSH directory entry and 10 PDF files. The titles include SSH Attack Simulation, SSH Detection Script, SSH Log Investigation, SSH service, UFW rules, Cron-Automation, Firewall-Configuration, Network configuration, IOC-Investigation, and Validation-and-Testing. The item titled SSH Detection Script is itself a PDF; I did not identify an actual script, private key, SSH configuration file, or raw SSH log in the reviewed folder export. The 10 exported PDFs are hash-matched to the supplied hash CSV.

### My pic and Our pic

I found no exact directory entries or path-specific content for My pic or Our pic in the reviewed Autopsy CSV or extracted [root] tree. The Autopsy file-type view shows zero image results, and its ingest-job table is empty. I therefore cannot infer that these folders were absent or unrecoverable; image-type analysis may not have run or may not be represented in the case outputs.

### ADDS Project

I found no exact ADDS Project directory entry in the reviewed deleted-file export or extracted root tree. The [orphan] tree contains PDFs with titles related to Active Directory structure and deployment, DHCP configuration, Group Policy, file services, Windows Server 2022 installation, Windows 10 domain joining, network design, NAT, Hyper-V, and security validation. A network design project folder also contains topology and switch-configuration images. These files may be relevant to an AD/network project, but their orphaned paths do not establish that they came from the target ADDS Project directory.

### Corel

The Corel directory is represented by an unallocated entry. The extracted tree contains 486 payload files, predominantly Corel application resources (presets, brush assets, fills, patterns, and image assets). Two CDR files appear under Corel Content\Images\Starter pack. I did not find support to characterize this entire folder as user-created design work; the path structure is consistent with bundled application content.

### Corel Draw Work

The corel draw work directory is represented by an unallocated entry. Its extracted folder contains 45 payload files, including three CDR files, 37 JPGs, two PNGs, two PDFs, and one ZIP. I verified their exported copies against File Hash Value.csv. The three CDR documents are potential design-project evidence, but I did not examine their contents or metadata in the source application, and I cannot identify an author or establish that the files were created by a specific person.

## 7. Timeline

I found no timestamp that directly records the deletion event. The dates below are file-system timestamps in the Autopsy export or image-acquisition times in the FTK summary. They establish context and acquisition sequence; they do not, by themselves, prove when a folder was deleted.

| Time (WAT)                       | Artifact / field                                              | What it supports                                                        | Does it prove deletion? |
|----------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------|
| 24 Sep 2026<br>12:39:46          | SSH directory Created Time in Autopsy CSV.                    | A creation-time value associated with the unallocated directory record. | No.                     |
| 24 Sep 2026<br>12:45:41          | SSH Modified/Change Time in Autopsy CSV.                      | Directory timestamp values; source field meaning should be preserved.   | No.                     |
| 24 Sep 2026<br>12:40:08–12:40:09 | Corel directory Create/Modified/Change values in Autopsy CSV. | Timestamp values associated with the unallocated record.                | No.                     |
| 24 Sep 2026<br>12:42:36          | corel draw work directory times in Autopsy CSV.               | Creation/modified/change values for the directory record.               | No.                     |
| 27 Sep 2026<br>16:30:25–16:31:06 | FTK acquisition interval in Omonoa Rerun.001.txt.             | When the image acquisition was recorded.                                | No.                     |
| 27 Sep 2026<br>16:31:06–16:31:13 | FTK verification interval.                                    | When the image verification was recorded.                               | No.                     |

The deleted-file CSV contains timestamps spanning 0000-00-00 through 24 Sep 2026 13:16:09 WAT. Zero dates are invalid/unknown values and should not be used as event times. The folder-entry times are not deletion times. I cannot give a defensible deletion timeframe from the current records; I would need correlated MFT/USN journal, recycle-bin, event-log, or other activity artifacts and a review of their timestamp semantics.

## 8. Cross-Tool and Export Validation

| Finding               | Comparison                                                                                                                                      | Assessment                                                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Omonoa image identity | FTK summary lists four segments and 5,008 MB source; Autopsy database records 5,251,285,504-byte Omonoa Rerun image and Africa/Lagos time zone. | Consistent identifiers and size support Autopsy Project 2 being based on the replacement set.       |
| Image integrity       | FTK summary records MD5 c798... and SHA-1 9951... matching report hashes; FTK verification screenshots show matching results.                   | Image bytes match the stored/report values; this does not restore the one unreadable source sector. |
| Deleted-file count    | Autopsy screenshot shows 1,162 deleted entries; CSV contains 1,162 rows.                                                                        | Internal consistency between Autopsy UI and export.                                                 |

| Finding                 | Comparison                                                                                                                           | Assessment                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| SSH exports             | Autopsy CSV lists 10 SSH PDFs as unallocated; [root] export contains 10 corresponding PDFs; all match File Hash Value.csv MD5/SHA-1. | Strong export consistency; hash-list image identity should still be documented.                         |
| Corel Draw Work exports | Autopsy CSV lists CDR/image/PDF/ZIP items under unallocated path; [root] export has 45 payload files; all match hash list.           | Strong export consistency; no author or deletion time established.                                      |
| R-Drive output          | R-Drive screenshot shows image creation completed; a .rdr file exists.                                                               | No verification/hash record ties the .rdr to the FTK/Autopsy image; no independent comparison possible. |
| Separate Evidence.E01   | FTK verify screenshot shows its own matching hashes; Autopsy screenshot shows failed ingest at sector offset 34.                     | Different sector count/hash from Omonoa set; keep separate until source relationship is documented.     |

These comparisons meet part of the assignment's cross-validation objective, but they do not establish that the separate Evidence.E01 hash list and the Omonoa Autopsy case represent the same acquisition. I cannot claim complete independent cross-tool validation until the image lineage is reconciled and the R-Drive output is verified.

## 9. Limitations

- The FTK acquisition summary records one unreadable source sector (10,256,384) replaced with zeros in the image.
- The assignment case ID AIT-DF-2026-009 differs from the FTK summary Case Number 27-09 and Evidence Number 09-26. The acquisition summary also names examiner Darego Omonoa Awoteinm; I have not independently confirmed the mapping or identity.
- The folder contains no actual Evidence.E01 image file, although screenshots and File Hash Value.csv refer to Evidence.E01. The earlier D:\Evidence.E01 copy examined was a 10.3 MB all-0xFF file; I cannot prove whether it is the same as, or was replaced by, the Evidence.E01 shown in later screenshots.
- Autopsy Project 2 CSV has no MD5 or SHA-256 values for its 1,162 deleted entries. The database has no ingest\_job records, and file-type categories display zero results; processing completeness is not established.
- The hash list refers to Evidence.E01, while Autopsy Project 2 refers to Omonoa Rerun.001. The folder does not contain an explicit source-to-source correlation or chain-of-custody record.
- The R-Drive output file exists, but its settings, image hash, and verification report are not present in the reviewed materials.
- No DD command/output was found. No chain-of-custody log, examiner processing log, evidence-transfer record, or authorization document was provided.

- No deletion timestamp, actor attribution, intent evidence, or file-copy-before-deletion evidence is established by these materials.

## 10. Conclusions

I conclude that the NIIT folder contains a replacement image set whose FTK acquisition report records successful MD5 and SHA-1 verification, subject to a single unreadable source sector zero-filled during acquisition. The Autopsy Project 2 case analyzes that Omonoa Rerun set and records 1,162 deleted-file entries. Three target directory names—SSH, Corel, and corel draw work—appear as unallocated entries. Exported content copies for those folders are present and match the corresponding MD5/SHA-1 entries in File Hash Value.csv.

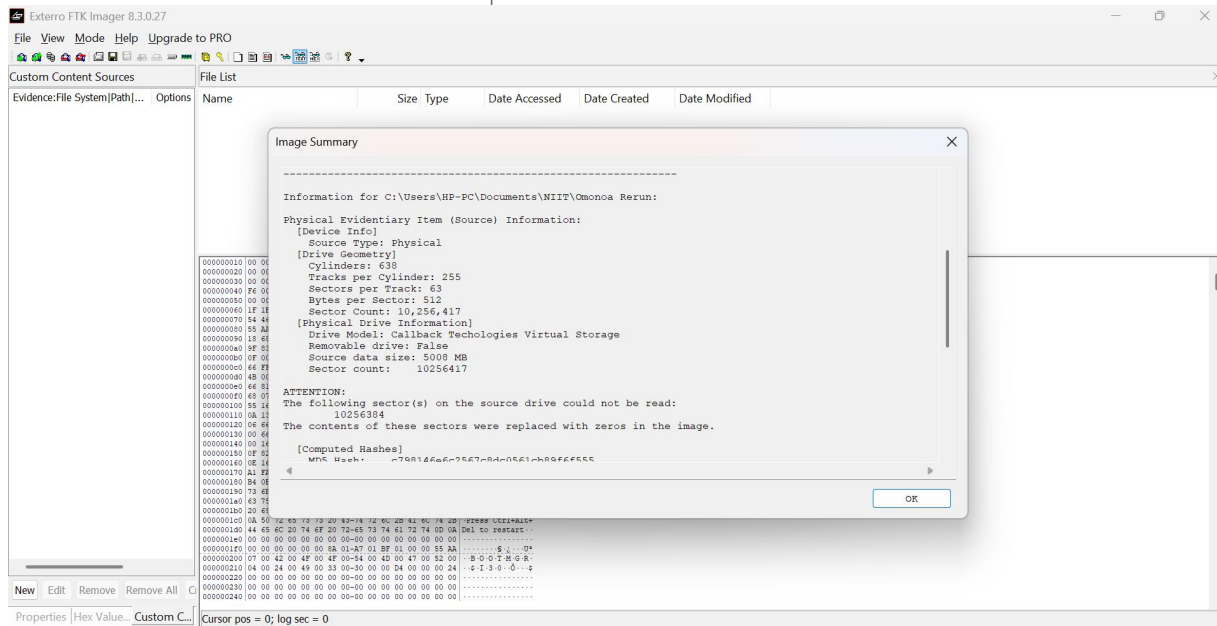
I cannot conclude that the other three target folders never existed, identify a deletion date, attribute the deletion to any person, or establish intent. I also cannot treat Evidence.E01 and Omonoa Rerun as one image without source lineage documentation. These are preliminary findings from supplied case artifacts, not a completed full forensic examination of all image sectors and artifacts.

## 11. Recommended Next Steps

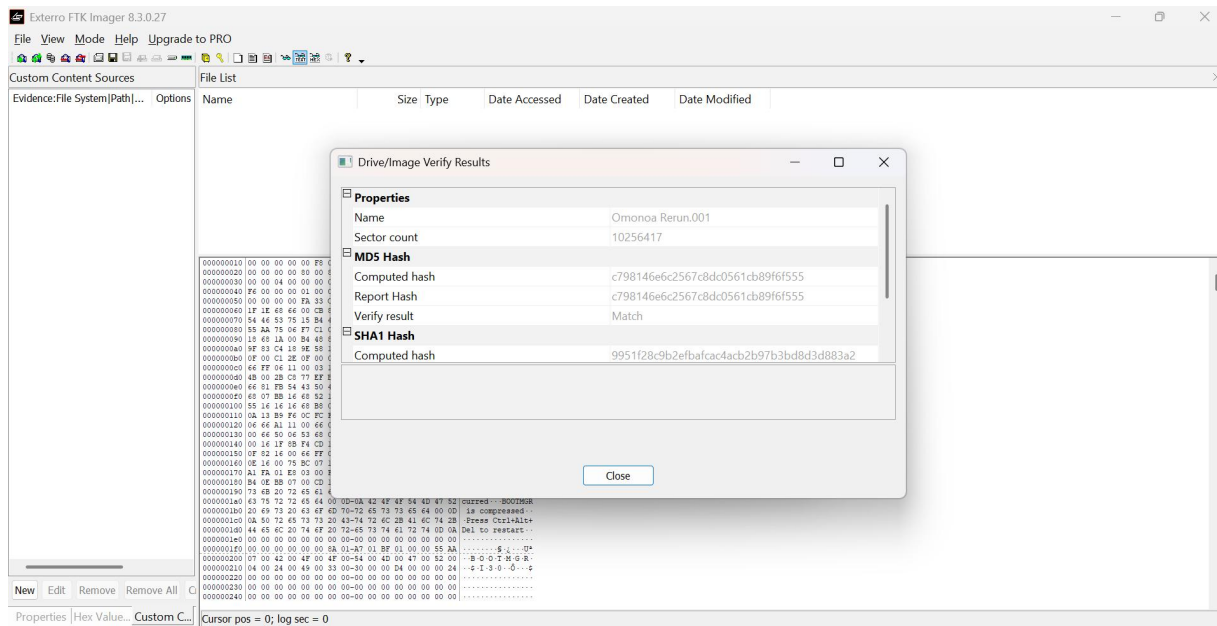
- Confirm that Omonoa Rerun.001–.004 is the replacement image sent by the instructor and map its FTK case/evidence identifiers to AIT-DF-2026-009. Retain the acquisition report with the evidence.
- Preserve the four image segments read-only and record fresh SHA-256 hashes for each segment and/or the full image set. Maintain a chain-of-custody and processing log.
- Run or document Autopsy ingest for the verified Omonoa image. Enable file-type identification, hash calculation, picture analysis, and relevant artifact/timeline modules; retain ingest logs.
- Export recovered target content to a separate evidence folder. For every recovered item record source path, status, recovery tool, metadata, SHA-256 hash, and screenshot. Keep the existing CSV hash matches as an integrity check.
- Investigate the 3 CDR files and related previews under corel draw work, and determine whether the orphaned AD/network documents can be tied to the ADDS Project by MFT, parent-directory, shortcut, or application artifacts.
- Reconcile the separate Evidence.E01 screenshot/hash list against the replacement image and the earlier all-0xFF copy before using its results in the case report.
- Build the deletion timeline from file-system and OS artifacts and state the limits of each timestamp. Do not use directory creation/change times as deletion times.
- If required by the assignment, document DD as unavailable/not applicable to the already acquired replacement image, with the instructor's acceptance or course requirements recorded.

## Appendix A. Selected Screenshots

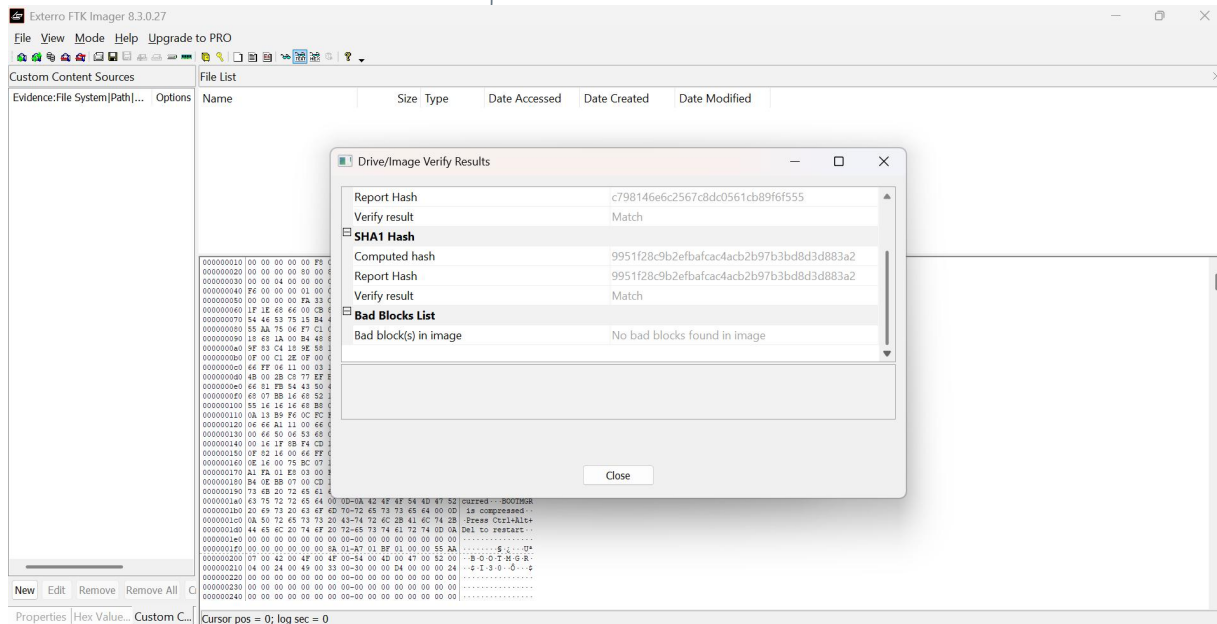
I selected these crops to document the replacement-image acquisition, verification, tool activity, and Autopsy results. The original screenshots remain unchanged in the NIIT folder. These are screen observations; where an image or case identity is uncertain, I state that limitation in the main report.



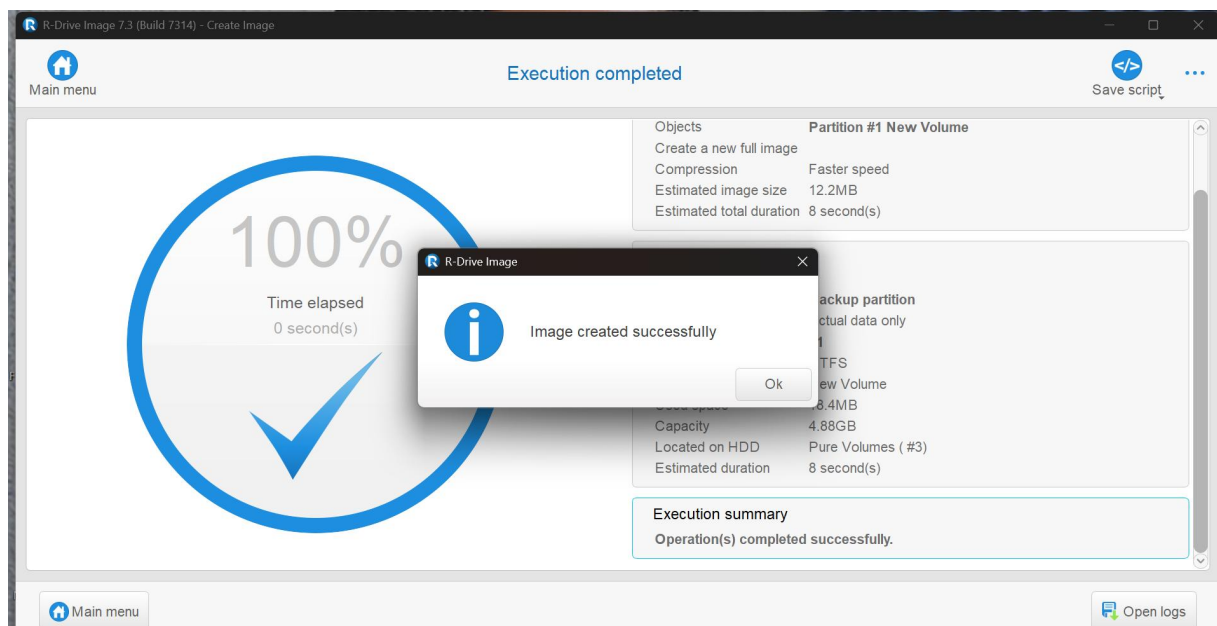
**Exhibit 52 | FTK Imager acquisition summary; physical source and one unreadable sector are visible.**



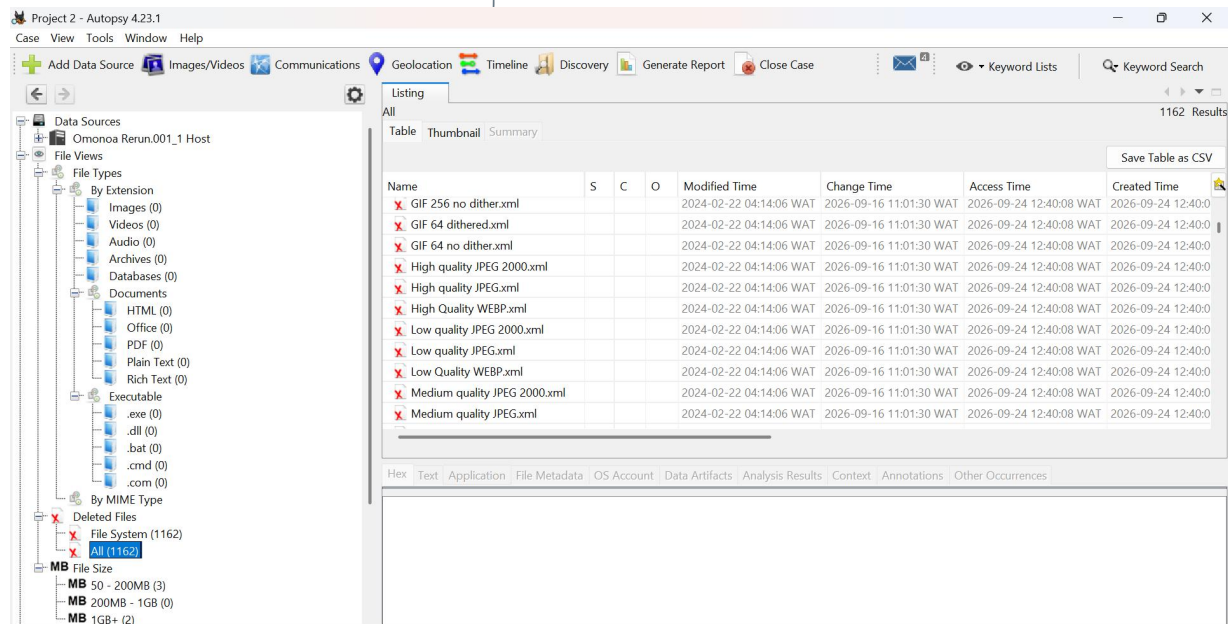
**Exhibit 48 | FTK verification for Omonoa Rerun.001; computed and report MD5 values match.**



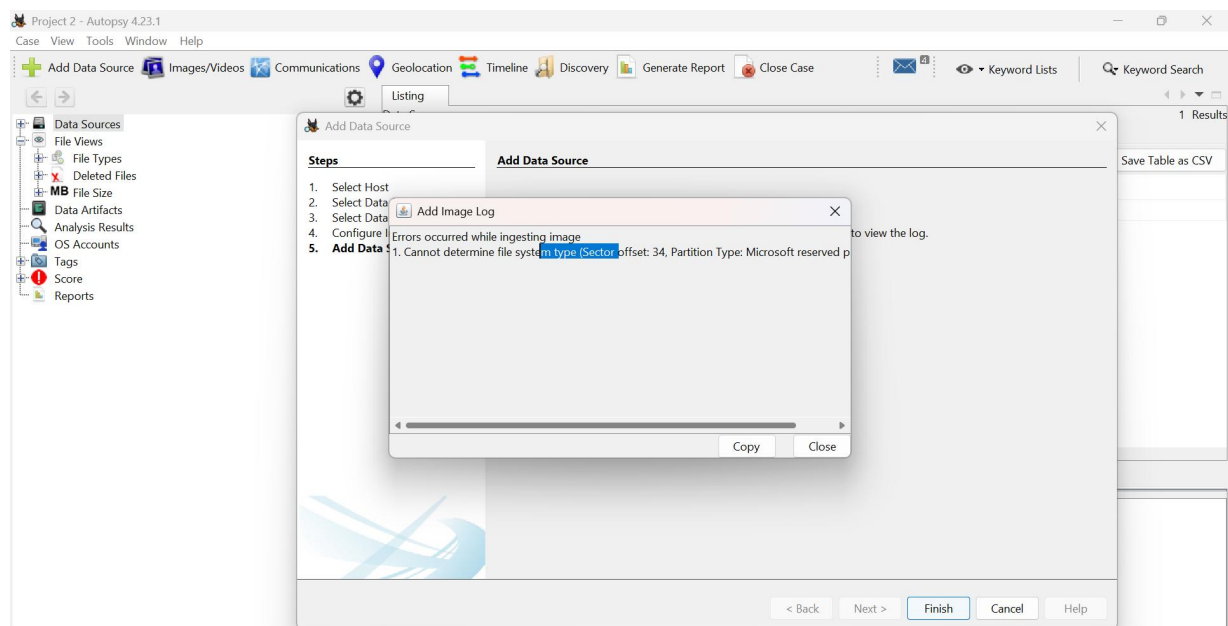
**Exhibit 49 | FTK verification for Omonoa Rerun.001; SHA-1 result and bad-block list.**



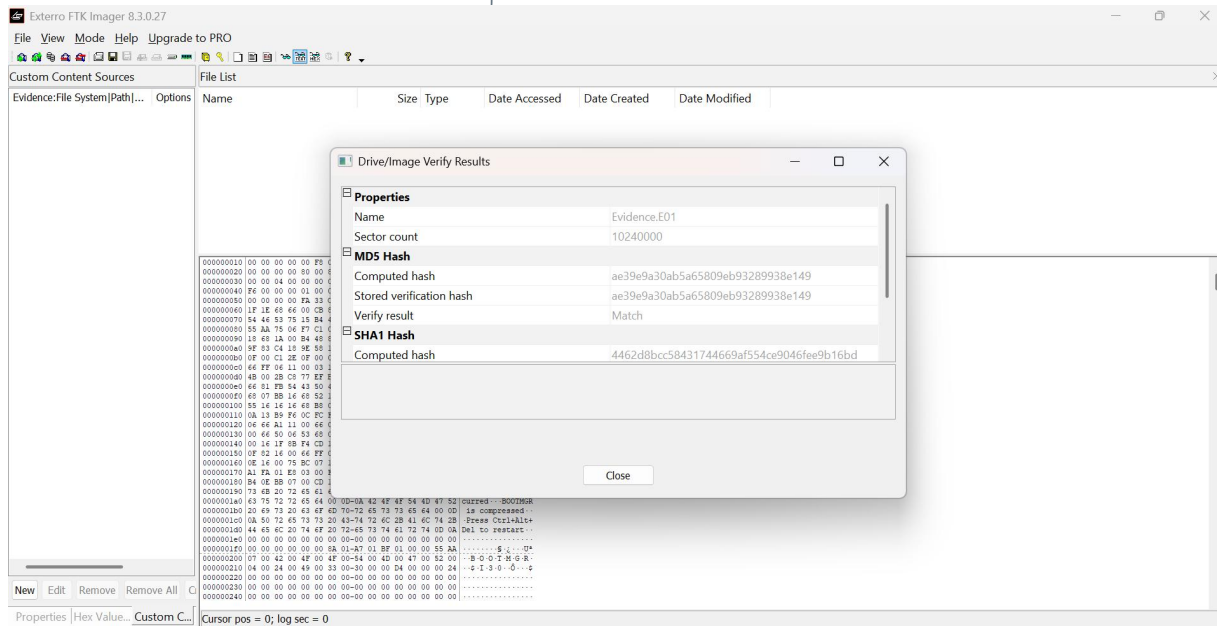
**Exhibit 57 | R-Drive Image reports image creation completed successfully.**



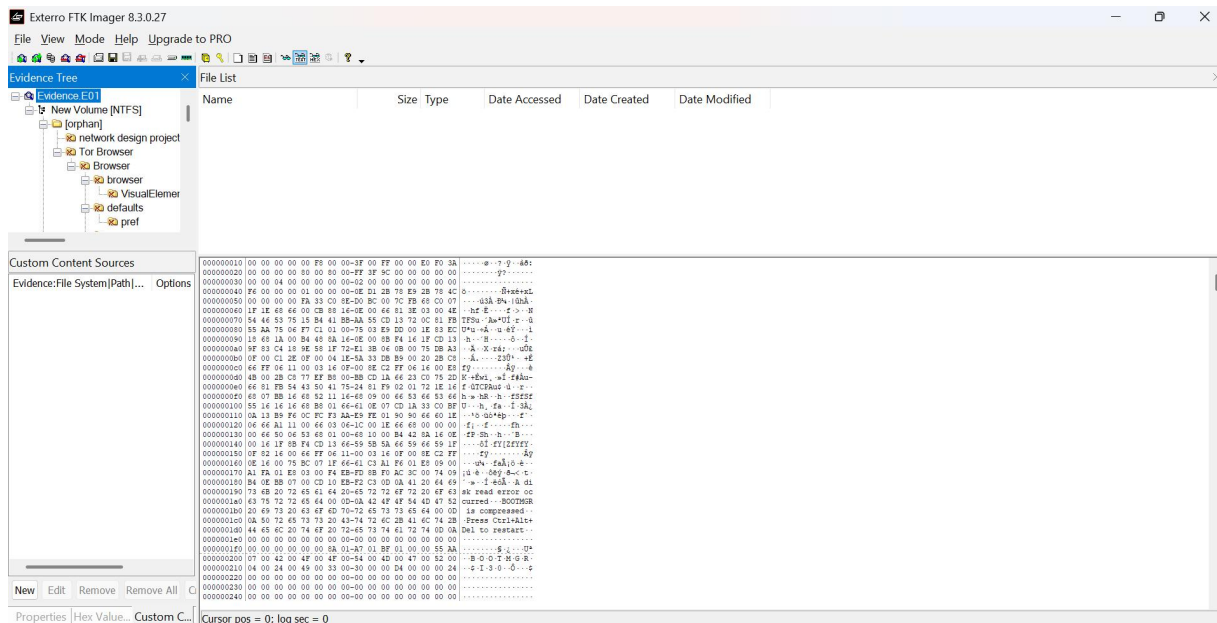
**Exhibit 60 | Autopsy Project 2 displays 1,162 deleted-file results.**



**Exhibit 58 | Autopsy reported an ingest error while adding the separate Evidence.E01 image.**



**Exhibit 43 | FTK verification for the separate Evidence.E01 reference; hashes differ from Omonoa Rerun.**



**Exhibit 62 | FTK Imager Evidence.E01 tree showing an [orphan] branch.**

## Appendix B. Evidence Registers

| Item                | Recorded value / status                           |
|---------------------|---------------------------------------------------|
| Primary image       | Omonoa Rerun.001–.004; total 5,251,285,504 bytes. |
| Primary image MD5   | c798146e6c2567c8dc0561cb89f6f555                  |
| Primary image SHA-1 | 9951f28c9b2efbafcac4acb2b97b3bd8d3d883a2          |

| Item                                   | Recorded value / status                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------|
| Unreadable source sector               | 10,256,384; zero-filled in image per acquisition summary.                                               |
| Separate Evidence.E01 screenshot MD5   | ae39e9a30ab5a65809eb93289938e149                                                                        |
| Separate Evidence.E01 screenshot SHA-1 | 4462d8bcc58431744669af554ce9046fee9b16bd                                                                |
| Recomputed exported-file hashes        | 541 files checked against File Hash Value.csv: SSH 10/10; Corel 486/486; corel draw work 45/45 matched. |
| Autopsy deleted-file CSV               | 1,162 rows; no MD5 or SHA-256 values populated.                                                         |
| R-Drive output                         | Partition3-1-image.rdr; 700,416 bytes; independent verification record not supplied.                    |
| DD                                     | No DD processing record found.                                                                          |
| Chain of custody                       | [PLACEHOLDER — add custodian, transfers, dates/times, signatures, and storage locations.]               |
| Investigator / team                    | [PLACEHOLDER — verify examiner name and team.]                                                          |

### Appendix C. Source Materials Reviewed

Case assignment: Forensic Project.docx (previously supplied). Replacement-image folder: C:\Users\HP-PC\Documents\NIIT. Primary image record: Omonoa Rerun.001.txt. Autopsy case: Project 2\autopsy.db and Project 2 CSV.csv. Hash register: File Hash Value.csv. Screenshots: Autopsy Screenshots\Screenshot (41).png through Screenshot (62).png. Exported trees: [root], [orphan], and [unallocated space].

The text and labels within the submitted case files and screenshots are treated as forensic source material. They do not alter the scope or conclusions of this report unless corroborated by the underlying image or another reliable record.